

LECTURE NOTES · COMPLETE SET

Advanced Exploratory Data Analysis

DATA209 · Written notes for all fifteen lecture pairs

PROGRAMME	BTech Hons. (Data Science), Semester III
SCHOOL	School of Engineering and Technology, Vidyashilp University
STRUCTURE	30 lecture hours · 60 laboratory hours · 15 weeks
CREDITS	4 (L:T:P = 2:0:4)
INSTRUCTOR	Dr. Shital Bhatt, Associate Professor
OUTCOMES	CO1–CO4

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Each set of notes lists the topics covered, the slide range in the module deck, the laboratory session that follows, key terms, common errors, free readings and self-check questions. Notes for L13-14 include Assignment 1 preparation; L29-30 closes with a whole-semester review.

LECTURE L1-2 • MODULE 1 • CO1

The role of EDA, and robust statistics

Week 1 | Slides: DATA209_Module_1_Statistical_and_Data_Foundations.pptx, pages 4-19 | Laboratory: P1-2

TOPICS COVERED

- Role of EDA in the data science lifecycle
- Exploratory vs. confirmatory analysis
- Sampling bias and inference intuition
- Distribution behaviour: skewness, kurtosis, outliers
- Summary and robust statistics — median, IQR

What EDA is for

Exploratory data analysis is the practice of looking at what the data *seems* to say before any hypothesis is imposed on it. Tukey named the field in 1977 as a deliberate counterweight to a statistical culture that had become almost entirely confirmatory. His argument was that the analyst's first duty is detective work — finding what is there — and that this work has its own methods, standards and failure modes.

The practical case is simpler. You cannot draw reliable conclusions from a large quantity of data by gleaning over it. Raw data is skewed, contains impossible values, and hides its own collection history. A model fitted to unexamined data inherits every one of those defects and reports them with false confidence.

Exploratory versus confirmatory

	EXPLORATORY	CONFIRMATORY
Question	open — what is going on here?	fixed before the data is seen
Hypothesis	an output	an input
Method	graphical, iterative, sceptical	formal tests, error rates
Success	a surprising, checkable claim	a defensible accept/reject decision

The cardinal error is switching modes silently: exploring until something looks significant, then reporting it as though the hypothesis came first. That is p-hacking, and it is the reason EDA findings are stated as *hypotheses for future testing* rather than as results.

Sampling and what you may claim

Every statistic is an estimate from a window onto a population. Four ways that window distorts:

- **Selection bias** — the sampling mechanism favours certain units. Website analytics only observe people who reached the website.
- **Survivorship bias** — failures are absent. Loan data holds approved applicants; rejected ones

are invisible.

- **Non-response bias** — those who did not answer differ systematically from those who did.
- **Measurement bias** — the instrument distorts. A sensor drifts; a zero means "not recorded".

Larger samples reduce variance, **not bias**. A biased frame stays biased at any size. Before generalising, ask the question that settles most disputes: *who could have been in this dataset but is not?*

Summary statistics: position, dispersion, shape

Three families of summary, and the discipline of pairing them correctly.

- **Position** — mean (balance point), median (splits the data in half), mode (most frequent).

The mean is not wrong on skewed data; it answers a different question. The size of the mean–median gap is the diagnostic.

- **Dispersion** — range, variance, standard deviation, IQR. The first three are computed around

the mean and inherit its sensitivity to extremes. The IQR is the width of the middle 50% and ignores the tails entirely.

- **Shape** — skewness measures horizontal asymmetry; kurtosis measures tail weight.

Report **mean with standard deviation**, or **median with IQR**. Never mix a mean with an IQR.

On `pima.csv`, Insulin has skew +2.27; its mean is 155.5 against a median of 125.0. Both numbers are correct. Only one describes a typical patient.

Reading a boxplot

The box spans Q1 to Q3 with the **median** — not the mean — inside it. Whiskers extend to the furthest point within $1.5 \times \text{IQR}$ of the box edges; points beyond are *flagged*, not condemned. Box position within the whiskers reveals skew: if $Q3 - Q2$ exceeds $Q2 - Q1$, the distribution leans right. A boxplot hides multimodality completely, so always pair it with a histogram or density plot.

KEY TERMS

Exploratory data analysis Confirmatory analysis Sampling frame Selection bias
Survivorship bias Robust statistic Skewness Kurtosis Interquartile range
Five-number summary

COMMON ERRORS TO AVOID

- Reporting a mean for a strongly skewed variable without also giving the median
- Pairing a mean with an IQR, or a median with a standard deviation
- Generalising beyond the sampling frame without saying so
- Treating flagged boxplot points as errors before investigating them

1. NIST/SEMATECH e-Handbook — Ch.1 Exploratory Data Analysis
<https://www.itl.nist.gov/div898/handbook/eda/eda.htm>
2. Downey — Think Stats, 2nd ed. (Ch. 1–4)
<https://greenteapress.com/wp/think-stats-2e/>
3. Adhikari & DeNero — Computational and Inferential Thinking
<https://inferentialthinking.com/>
4. MIT OCW 6.0002 — sampling and statistical fallacies
<https://ocw.mit.edu/courses/6-0002-introduction-to-computational-thinking-and-data-science-fall-2016/>

TEST YOURSELF

1. Give a variable for which the mode is the only meaningful measure of position.
2. A dataset has mean 40 and median 12. Sketch the distribution and name the summary you would report.
3. Why does increasing the sample size not fix selection bias?
4. $Q_3 - Q_2 = 5$ and $Q_2 - Q_1 = 18$. Which way does the distribution lean?

LECTURE L3-4 • MODULE 1 • CO1

Data types and representation

Week 2 | Slides: DATA209_Module_1_Statistical_and_Data_Foundations.pptx, pages 20–30 | Laboratory: P3-4

TOPICS COVERED

- Nominal, ordinal, interval and ratio scales
- Discrete vs. continuous
- Structured vs. semi-structured vs. unstructured data
- Text representation basics
- Time-series behaviour: trend and seasonality

Measurement scale decides the statistic

The dtype pandas infers is a storage decision. The **measurement scale** is a semantic one, and it governs which operations are meaningful.

SCALE	PROPERTY	LEGAL	ILLEGAL
Nominal	names, no order	counts, mode, chi-square	mean, median, ordering
Ordinal	ordered, unequal gaps	median, percentiles, rank tests	mean (debatable), differences
Interval	equal gaps, arbitrary zero	mean, standard deviation, differences	ratios
Ratio	equal gaps, true zero	everything, including "twice as much"	—

An integer column can be nominal. In the shoppers dataset `OperatingSystems`, `Browser`, `Region` and `TrafficType` are stored as integers and are pure codes; averaging them produces a number with no referent. This is the single most common error in submitted assignments.

Independently of scale, variables are **discrete** (countable — page views, pregnancies) or **continuous** (any value in a range — BMI, duration). Discrete versus continuous decides the plot; the measurement scale decides the statistic.

Structure

- **Structured** — fixed schema, rows and columns. CSV, SQL, spreadsheets. Directly analysable.
- **Semi-structured** — self-describing but irregular. JSON, XML, logs. Needs flattening first.
- **Unstructured** — no tabular form. Text, images, audio, video. Requires a representation

decision before any analysis at all.

Storage format matters too: CSV carries no dtypes and breaks on delimiters inside text; Excel silently coerces types and hides rows; Parquet is typed and columnar but not human-readable.

Text becomes countable

Raw text is human-readable and machine-useless. Exploration begins after a representation choice:

1. **Tokenise** — split into words; normalise case, strip punctuation, optionally drop stop words.
2. **Bag-of-words** — count each vocabulary term per document. Interpretable; ignores word order.
3. **TF-IDF** — down-weight terms common across all documents, up-weight distinctive ones.
4. **N-grams** capture short phrases; **embeddings** capture meaning — both cost interpretability.

For exploratory purposes, document-length distribution, vocabulary size and ranked term frequency are usually sufficient. A ranked frequency bar chart is precise; a word cloud is decorative.

Time series

Four components are present in every series: **trend** (long-run direction), **seasonality** (a repeating pattern of fixed period), **cyclic** behaviour (long waves of no fixed period) and the **irregular** residual.

Two rules follow, and both matter for the final project: with time-ordered data you may **not shuffle rows**, and you may **not split randomly**. Training on rows recorded after the test rows is temporal leakage, covered properly in L29-30.

An ordered **Categorical** for a month column is not cosmetic. Sorted alphabetically, a seasonal pattern becomes invisible; sorted chronologically, the same data shows it immediately.

KEY TERMS

Nominal	Ordinal	Interval	Ratio	Discrete	Continuous	Semi-structured data
Tokenisation	Bag-of-words	TF-IDF	Trend	Seasonality		

COMMON ERRORS TO AVOID

- Averaging an integer column that encodes a nominal category
- Leaving a month or size column unordered, so plots sort alphabetically
- Counting categories before trimming whitespace and unifying case
- Splitting time-ordered data randomly

READINGS — ALL FREE TO ACCESS

1. pandas user guide — categorical data
https://pandas.pydata.org/docs/user_guide/categorical.html
2. pandas user guide — time series
https://pandas.pydata.org/docs/user_guide/timeseries.html
3. McKinney — Python for Data Analysis, 3rd ed.
<https://wesmckinney.com/book/>
4. Kaggle Learn — Pandas
<https://www.kaggle.com/learn/pandas>

TEST YOURSELF

1. Postal codes are integers. What scale are they, and what may you compute?
2. Why is the mean of a five-point Likert scale contested?
3. Name a variable that is discrete and ratio-scaled.
4. Give one reason a random train/test split invalidates a time-series analysis.

LECTURE L5-6 • MODULE 1 • CO1

Unstructured data and ethics

Week 3 | Slides: DATA209_Module_1_Statistical_and_Data_Foundations.pptx, pages 31–37 | Laboratory: P5-6

TOPICS COVERED

- Image as a matrix; audio as a waveform
- Video as a sequence of frames
- Curse of dimensionality — introduction
- Data bias and ethical concerns

Everything becomes an array

- **Image** — a grid of pixel values. Grayscale is `(height, width)`; colour is

`(height, width, 3)`. Explore the intensity histogram: exposure, contrast, and clipping at 0 or 255 tell you what was lost at capture and cannot be recovered.

- **Audio** — amplitude sampled over time: a 1-D array plus a **sample rate**. Explore the

waveform for structure, the amplitude histogram for symmetry, and short-term energy for loudness over time. Silence is often the majority of a field recording.

- **Video** — a sequence of frames, `(frames, height, width, channels)` plus a frame rate. Each

frame is an image, so image technique applies frame by frame.

Once represented as an array, the ordinary exploratory questions apply unchanged: what is the distribution, where are the extremes, what is missing, and is anything clipped or saturated?

The curse of dimensionality — a flag planted

As dimensions grow, data density collapses exponentially. In a unit cube, the proportion of volume lying within 0.1 of an edge is 20% in one dimension, 49% in three, and 99% in twenty — almost every point is near a boundary. Distances between points converge, so "nearest neighbour" stops carrying information, and distance-based methods (k-NN, k-means) degrade first.

This is introduced here only. The response — feature selection and dimensionality reduction — is Module 4.

Bias enters before you write any code

Four kinds, all present before analysis begins:

- **Historical bias** — the world the data records was already unequal. Faithful data can encode unfair outcomes.
- **Representation bias** — some groups are under-sampled; findings will not transfer to them.

- **Measurement bias** — the proxy is not the thing. Arrests are not crime; clicks are not interest.
- **Aggregation bias** — one model for everyone hides subgroup differences. Simpson's paradox is

the statistical form of this, and it appears again in L11-12.

An ethics checklist

Run this before analysing, not after:

1. Who is in this dataset, who is missing, and would they have consented to this use?
2. Does it contain personal or sensitive attributes, and do you actually need them?
3. Could a finding here be used to disadvantage a group represented in the data?
4. Are proxy variables standing in for protected attributes?
5. What claim would be unsupportable outside the sampling frame — and have you written it down?

The Pima dataset records women of Pima heritage aged 21 and over. Every conclusion drawn from it is conditional on that frame, and a model trained on it must not be presented as a general screener.

KEY TERMS

Modality

Sample rate

Pixel intensity

Clipping

Curse of dimensionality

Historical bias

Representation bias

Proxy variable

Aggregation bias

COMMON ERRORS TO AVOID

- Reporting image statistics without stating shape, dtype and value range
- Treating a proxy variable as if it were the quantity of interest
- Presenting a model trained on one population as generally applicable
- Leaving the ethics review to the end of the project

READINGS — ALL FREE TO ACCESS

1. scikit-image — images as NumPy arrays
https://scikit-image.org/docs/stable/user_guide/numpy_images.html
2. Wilke — Fundamentals of Data Visualization
<https://clauswilke.com/dataviz/>
3. Berkeley Data 100 — course notes
<https://ds100.org/course-notes/>

TEST YOURSELF

1. An 8-bit grayscale image has 12% of pixels at exactly 255. What happened, and can it be undone?
2. Why do distance-based methods degrade first in high dimensions?
3. Give an example of a proxy variable that could stand in for a protected attribute.
4. State one claim you could not defend from the Pima dataset.

LECTURE L7-8 • MODULE 2 • CO2

Univariate EDA

Week 4 | Slides: DATA209_Module_2_Core_Advanced_EDA.pptx, pages 3-10 | Laboratory: P7-8

TOPICS COVERED

- Numerical variables: boxplot, density, distribution comparison
- Categorical variables: frequency tables, bar plots
- Detecting skewness and anomalies

Two families, used in order

Non-graphical technique produces numerical summaries — quantiles, moments, frequency and contingency tables, correlation matrices. It is precise, reproducible and easy to tabulate, and it is blind to shape. Anscombe's quartet is four datasets sharing means, variances and correlation coefficients while looking completely different.

Graphical technique reveals shape, gaps, clusters and outliers instantly, and is sensitive to choices you make — bin width, axis limits, smoothing bandwidth.

Use both, in this order: compute the summary, plot it, then reconcile the two when they disagree.

The four questions

Answer all four for every column before looking at a single relationship.

1. **What is the centre?** Mean or median, whichever the skew justifies. State which you used.
2. **What is the spread?** Standard deviation or IQR, plus the range — which exposes impossible values.
3. **What is the shape?** Symmetric, skewed, bimodal, uniform, zero-inflated. This decides everything downstream.
4. **What is unusual?** Outliers, gaps, spikes at round numbers, impossible values.

Choosing a distribution plot

PLOT	SHOWS	WEAKNESS
Histogram	frequency per bin	bin width changes the story
Density (KDE)	smoothed shape	bandwidth invents smoothness; spills past data limits
Boxplot	five-number summary	hides multimodality completely
Violin	density plus summary	cluttered with many groups
ECDF	cumulative proportion	less intuitive to a general audience

Default to histogram plus boxplot. Add a density overlay when comparing groups; reach for the ECDF when bin width is contested, since it requires no binning choice at all.

Reading anomalies off a plot

- A **spike at exactly zero** usually means "not recorded", not "measured as zero".
- **Two humps** mean two populations are mixed — find the variable that separates them.
- A **hard wall** at a round number suggests a cap, truncation or data-entry limit.
- **Spikes at multiples of 5 or 10** indicate self-reported or rounded data.
- A **long thin tail** is skew; **isolated far points** are candidate outliers. Different problems.

In `pima.csv` this reading finds 374 zeros in Insulin and 227 in SkinThickness — every one a disguised missing value that `isna()` cannot see. Module 3 treats them; Module 2 finds them.

Categorical univariate

Frequency tables with counts *and* proportions. Check cardinality (5 levels is a bar plot, 500 is a grouping problem) and imbalance. The shoppers `Revenue` flag is roughly 84.5% / 15.5%; the Pima outcome is 65% / 35%. Record this now — imbalance discovered here changes your sampling and evaluation strategy in Module 4, where a model predicting "never" would score 84.5% accuracy.

KEY TERMS

Non-graphical EDA	Graphical EDA	Anscombe's quartet	Kernel density estimate	ECDF
Zero-inflation	Multimodality	Class imbalance		

COMMON ERRORS TO AVOID

- Presenting a single histogram without checking a second bin width
- Trusting a KDE that extends past the physical limits of the variable
- Using a boxplot alone on data that turns out to be bimodal
- Failing to record class imbalance during univariate analysis

READINGS — ALL FREE TO ACCESS

1. seaborn — visualising distributions
<https://seaborn.pydata.org/tutorial/distributions.html>
2. NIST/SEMATECH — graphical techniques
<https://www.itl.nist.gov/div898/handbook/eda/eda.htm>
3. Berkeley Data 100 — EDA notes
<https://ds100.org/course-notes/>

TEST YOURSELF

1. Two variables share a mean and standard deviation. Name two plots that would distinguish them.
2. A rate variable's KDE shows density below zero. What went wrong?
3. Why is a spike at exactly zero worth more attention than a point in the far tail?
4. At 15% positive class, what accuracy does a model that never predicts positive achieve?

LECTURE L9-10 • MODULE 2 • CO2

Bivariate relationships

Week 5 | Slides: DATA209_Module_2_Core_Advanced_EDA.pptx, pages 11-20 | Laboratory: P9-10

TOPICS COVERED

- Correlation and covariance
- Scatterplots and what they hide
- Relationships between categorical variables

Covariance and correlation

Covariance is the average product of paired deviations from the two means. Its sign gives the direction; its magnitude is uninterpretable because it carries the units of both variables. Changing metres to centimetres changes the number without changing the relationship.

Correlation is covariance divided by the two standard deviations: bounded in $[-1, 1]$, unitless, and therefore comparable across pairs. That comparability is the entire point.

Choosing a coefficient

COEFFICIENT	MEASURES	USE WHEN	ROBUST
Pearson r	linear association	both continuous, roughly linear	no
Spearman ρ	monotonic association on ranks	non-linear but consistently increasing; ordinal	yes
Kendall τ	concordance of pairs	small samples; many ties	yes
Cramér's V	association between categoricals	two nominal variables	n/a
Correlation ratio η	categorical vs numeric	group means differ across a category	partly

Comparing Pearson with Spearman is a cheap non-linearity test: a large gap means the relationship is monotonic but not straight.

Correlation is not causation, and not shape

Correlation measures co-movement. It says nothing about which variable moved first, or whether a third moved both. And it says nothing about shape — hence the rule:

A correlation coefficient reported without its scatterplot is an unverified claim.

Five patterns worth naming when you look: **linear**, **non-linear** (Pearson understates it), **heteroscedastic** (spread widens along the axis), **clustered** (a single correlation over all points is meaningless — split first), and **outlier-driven** (a few points create or destroy the association; recompute without them and compare).

Relationships involving categories

- **Numeric vs categorical** — grouped boxplots or violins; compare medians *and* spreads, not just means. Quantify with a group-means table.

- **Categorical vs categorical** — cross-tabulation with **row or column percentages**, never raw counts. Quantify with chi-square, then Cramér's V for strength.

Percentages beat counts: 20 purchases from 40 sessions beats 100 from 1,000, and raw counts hide that. Always report the denominator alongside the rate.

With 12,330 rows almost every chi-square test returns a significant p-value. Read the **effect size**, not the p-value.

From pattern to hypothesis

A hypothesis names a **mechanism**, predicts a **direction**, and states what would **falsify** it.

- Weak: "Glucose is related to the outcome." This restates the correlation and predicts nothing.
- Better: "Elevated fasting glucose precedes diagnosis, so glucose will separate the classes better

than BMI — falsified if the separation vanishes after conditioning on age."

Name the confounders you cannot rule out with this data. Age confounds almost everything in the Pima dataset; the promotional calendar is unobserved in the shoppers dataset.

KEY TERMS

Covariance	Pearson correlation	Spearman correlation	Kendall tau	Cramér's V
Heteroscedasticity	Cross-tabulation	Chi-square test	Effect size	Confounder

COMMON ERRORS TO AVOID

-
- Running Pearson correlation on a nominal or ordinal column
 - Reporting a correlation without plotting the pair
 - Comparing raw counts across groups of very different sizes
 - Treating a small p-value on 12,000 rows as evidence of a large effect

READINGS — ALL FREE TO ACCESS

-
1. scikit-learn — clustering and metrics
<https://scikit-learn.org/stable/modules/clustering.html>
 2. seaborn — relational and categorical plots
<https://seaborn.pydata.org/tutorial.html>
 3. James et al. — Introduction to Statistical Learning
<https://www.statlearning.com/>

TEST YOURSELF

1. Pearson $r = 0.15$ and Spearman $\rho = 0.61$ for the same pair. What does that tell you?
2. Why is covariance rarely reported directly?
3. Two categories show conversion rates of 50% and 10%. What must you also report?
4. Rewrite 'sales correlate with advertising' as a falsifiable hypothesis.

LECTURE L11-12 • MODULE 2 • CO2

Multivariate structure

Week 6 | Slides: DATA209_Module_2_Core_Advanced_EDA.pptx, pages 21-29 | Laboratory: P11-12

TOPICS COVERED

- Multicollinearity and the variance inflation factor
- Identifying groups in data
- K-means for structure discovery
- Interpreting clusters

Why go past two variables

A relationship that holds within every subgroup can **reverse** when the subgroups are pooled. That is Simpson's paradox, and it is the strongest argument for multivariate analysis: two variables at a time can be consistently, confidently wrong.

Multicollinearity

Multicollinearity is strong correlation *among the predictors themselves*, not with the target. Its consequences:

- Coefficient estimates become unstable — small data changes swing them, sometimes flipping sign.
- Standard errors inflate, so genuinely useful variables appear insignificant.
- Predictive accuracy may be unharmed; **interpretability is destroyed**.

It is invisible in a target-correlation ranking. You must inspect the predictor-to-predictor block of the correlation matrix, then confirm with the **variance inflation factor**.

VIF for a predictor is $1 / (1 - R^2)$ from regressing that predictor on all the others. It answers: how redundant is this column?

VIF	READING	ACTION
≈ 1	uncorrelated with the others	keep
1 – 5	moderate	keep, note it
5 – 10	coefficients becoming unreliable	investigate the pair
> 10	largely reconstructable from the others	drop, combine, or reduce

In the shoppers data, **BounceRates** and **ExitRates** measure nearly the same thing. Keep one, or combine them — a decision revisited in L25-26.

K-means, exploratorily

Four steps: choose k and initialise (k-means++ rather than at random); **assign** each point to its nearest centroid; **update** each centroid to the mean of its members; **repeat** until assignments stop changing.

Scale the features first. K-means minimises Euclidean distance, so an unscaled variable with a large range decides the clusters by itself.

Choosing k , and being honest about the answer

The **elbow** on the inertia curve and the **mean silhouette** are the two standard aids. Silhouette compares each point's distance to its own cluster against the nearest other cluster, running from -1 to $+1$:

SCORE	READING
> 0.7	strong, well-separated
$0.5 - 0.7$	reasonable structure
$0.25 - 0.5$	weak; clusters overlap
< 0.25	no substantial structure

On standardised `pima.csv` the best score across $k = 2 \dots 6$ is about 0.197. The correct conclusion is that this dataset has no strong natural grouping — and reporting that honestly is a good answer. Manufacturing clusters that are not there is not.

Interpreting clusters

A cluster is only real if you can describe it. Profile each one by computing every feature's mean within it and comparing to the overall mean; name it in domain language ("older, higher BMI, high glucose"), not as "cluster 2". Check sizes — a cluster of four points is an outlier group, not a segment. Re-run with a different seed and a different k : structure that vanishes was never there.

KEY TERMS

Simpson's paradox	Multicollinearity	Variance inflation factor	k-means	k-means++
Inertia	Elbow method	Silhouette score	Cluster profiling	

COMMON ERRORS TO AVOID

- Running k-means on unscaled features
- Reporting clusters without profiling or naming them
- Choosing k from the elbow alone without a silhouette check
- Presenting weak clustering as clean market segments

1. **scikit-learn — clustering**

<https://scikit-learn.org/stable/modules/clustering.html>

2. **statsmodels — variance inflation factor**

https://www.statsmodels.org/stable/generated/statsmodels.stats.outliers_influence.variance_inflation_factor.html

3. **Hastie et al. — Elements of Statistical Learning, Ch. 14**

<https://hastie.su.domains/ElemStatLearn/>

TEST YOURSELF

1. A predictor has VIF 14. Name three possible responses.
2. Why can multicollinearity leave predictive accuracy intact while ruining interpretation?
3. Mean silhouette is 0.19 at every k you tried. What do you report?
4. Give a concrete example of Simpson's paradox.

LECTURE L13-14 • MODULE 2 • CO2

Visual storytelling

Week 7 | Slides: DATA209_Module_2_Core_Advanced_EDA.pptx, pages 30-36 | Laboratory: P13-14 (Assignment 1)

TOPICS COVERED

- Grammar of graphics
- Good vs. misleading plots
- Visualising non-numeric data
- Dashboard logic and storytelling with data

The grammar of graphics

A plot is built from layers. Thinking in layers is what lets you answer "which plot?" by reasoning rather than by recall.

1. **Data** — the tidy table being shown, one row per observation.
2. **Aesthetics** — the mapping from variables to visual properties: x, y, colour, size, shape.
3. **Geometries** — the marks that carry the data: points, bars, lines, boxes, tiles.
4. **Scales** — how data values become visual values: axis limits, log scales, colour palettes.
5. **Facets** — small multiples; the same plot repeated across the levels of a variable.
6. **Coordinates and theme** — coordinate system plus the non-data ink: labels, grid, legend.

Choosing a chart

YOU HAVE	QUESTION	PLOT
one numeric	what is its shape?	histogram, density, boxplot, ECDF
one categorical	how common is each level?	bar plot of counts or proportions
two numeric	do they move together?	scatterplot, plus a smoother
numeric + categorical	do the groups differ?	grouped boxplot, violin, overlaid densities
two categorical	are they associated?	cross-tab, stacked bar of proportions, mosaic
many numeric	where is the structure?	correlation heatmap, pair plot, parallel coordinates
numeric over time	what is the trend?	line chart, ordered by time

Pie charts appear nowhere in this table. They ask the reader to compare angles, which people do poorly; a bar chart beats them in almost every case.

Six ways a plot misleads

- **Truncated axis** — a bar chart not starting at zero exaggerates small differences.

- **Dual axes** — two scales on one chart manufacture correlation by choice of range.
- **3-D and area encodings** — doubling a radius quadruples the perceived size.
- **Cherry-picked range** — where the time axis starts decides whether the trend rises or falls.
- **Rainbow colour maps** — non-perceptual palettes create bands that are artefacts of the colours.
- **Hidden aggregation** — plotting means without spread or sample size conceals both.

The test to apply to your own work: *could someone reach the opposite conclusion from this data with a different but equally reasonable choice?*

Non-numeric data

Ordered bar plots for ordinal data — never let the software sort alphabetically. Stacked bars of proportions for part-to-whole comparisons. Mosaic plots when both variables are categorical with several levels. Heatmaps for contingency tables, using a perceptually uniform colour scale. For high cardinality, plot the top 15 and aggregate the rest into a labelled "Other".

Assignment 1 preparation

The mini-project due this week is marked on four criteria: **technique 30%**, **interpretation 35%**, **reproducibility 15%**, **presentation 20%**. Interpretation carries the most weight, which is deliberate — it is the skill the course exists to build.

What to bring to the presentation:

1. The problem statement and what a row represents, with the sampling frame stated.
2. The dependent/independent split, justified.
3. Univariate findings: shape, centre, spread and anomalies per variable.
4. Bivariate findings, including at least one relationship you did not expect.
5. Multivariate structure — grouping or clustering, interpreted in plain language.
6. Three hypotheses, each with a falsification condition and its confounders named.

Two failures cost the most marks: **figures with no written interpretation**, and **absolute file paths** that only run on your own machine. Restart the kernel and run all before submitting.

Narrative

Lead with the finding, not the method: "Glucose separates the classes" beats "we plotted glucose". One figure, one message — if a plot needs two sentences to explain, it is two plots. Sequence the report: what the data is → what is wrong with it → what it shows → what you cannot conclude. Annotate directly on the figure. State limitations in the same breath as the finding.

A **dashboard** answers recurring questions; a **report** answers one question well. Know which you are building before you start.

KEY TERMS

Grammar of graphics	Aesthetic mapping	Geometry	Facet	Small multiples	Truncated axis
Perceptually uniform colour map	Data-ink ratio				

COMMON ERRORS TO AVOID

- Truncating a bar chart's y-axis
- Using a pie chart to compare more than three categories
- Leaving a legend to do work that direct annotation would do better
- Publishing a figure with no sentence stating what it shows

READINGS — ALL FREE TO ACCESS

1. Wilke — Fundamentals of Data Visualization
<https://clauswilke.com/dataviz/>
2. CMU 36-315 — Statistical Graphics and Visualization
<https://cmu-36315.netlify.app/>
3. seaborn tutorial
<https://seaborn.pydata.org/tutorial.html>

TEST YOURSELF

1. Name the six layers of the grammar of graphics.
2. When is a truncated y-axis defensible?
3. You have two categorical variables with six levels each. Which plot, and why?
4. Rewrite 'Figure 3: PageValues by Revenue' as a title that states the finding.

LECTURE L15-16 • MODULE 3 • CO3

Data quality dimensions

Week 8 | Slides: DATA209_Module_3_Data_Quality_Engineering.pptx, pages 4-11 | Laboratory: P15-16

TOPICS COVERED

- Accuracy, completeness, consistency, validity
- Integrating data from multiple sources
- Duplicate detection, inconsistent encoding, validation rules

Cleaning is inference

Every cleaning decision is a modelling decision made early, with less evidence and usually without a record.

Imputing a mean asserts something about the missing values. Deleting an outlier asserts it was an error. Neither is neutral, and both must be written down.

Six dimensions

DIMENSION	QUESTION
Accuracy	do the values reflect reality? A blood pressure of 0 is precise, recorded and wrong
Completeness	is anything absent that should be present — including <i>disguised</i> absence?
Consistency	do values agree across records and sources?
Validity	do values conform to type, range, allowed set and cross-field rules?
Uniqueness	is each real-world entity represented once?
Timeliness	is the data current enough, and is every record from the same period?

Audit in this order: Validity failures often explain completeness failures, which often explain accuracy failures.

Profiling

Per column: dtype, nulls, **zeros**, unique count, min, max, top values. Then exact duplicate rows, then duplicates on the business key alone. For numeric columns check the range against what is physically possible; for categorical columns list the full value set and hunt for near-duplicates; for dates check range, ordering and parse failures.

The zeros column is the one that finds disguised missingness — see L17-18.

Multi-source integration

PROBLEM	SYMPTOM	RESPONSE
key mismatch	far fewer rows after the join	normalise keys: trim, case-fold, check type
granularity mismatch	row count explodes	aggregate the finer side first
schema drift	same name, different meaning or unit	rename explicitly; convert units
overlapping records	the same entity in both sources	deduplicate after the join
conflicting values	sources disagree	define precedence in advance; record which won

Always compare row counts before and after a join. **An unexplained change in row count is a bug until proven otherwise.**

Duplicates and encodings

Duplicates come in three grades: **exact** (identical across all columns), **key** (same entity, differing fields — the harder and more common case), and **fuzzy** ("Ramesh Kumar" vs "Ramesh Kumar", needing string distance). Decide which record survives: most recent, most complete, or a merge.

Inconsistent encodings: case and whitespace; mixed units in one column; DD/MM/YYYY against MM/DD/YYYY, silently wrong for the first twelve days of a month; sentinels such as 999, -1, NA, N/A, '' and unknown all meaning missing. **Standardise before you count** — every frequency table computed on unstandardised categories is wrong.

Validation rules

Write them from domain knowledge *before* looking at the data, then measure the violations. Type rules (age is an integer), range rules (glucose between 40 and 400), set rules (VisitorType is one of three known values) and cross-field rules (discharge is not before admission; a duration cannot be positive when the page count is zero).

The **cleaning log** — finding, rows affected, action, justification — is a deliverable, not a courtesy. It carries most of the marks in Assignment 2.

KEY TERMS

Data profiling	Accuracy	Completeness	Consistency	Validity	Uniqueness	Timeliness
Business key	Fuzzy duplicate	Sentinel value	Cleaning log			

COMMON ERRORS TO AVOID

- Joining two sources without comparing row counts before and after
- Counting categories before standardising case and whitespace
- Deriving validation rules from the data instead of from the domain
- Cleaning without recording what was changed and why

1. pandas — merge, join and concatenate
https://pandas.pydata.org/docs/user_guide/merging.html
2. Kaggle Learn — Data Cleaning
<https://www.kaggle.com/learn/data-cleaning>
3. Berkeley Data 100 — data cleaning notes
<https://ds100.org/course-notes/>

TEST YOURSELF

1. Your join turns 10,000 rows into 41,000. What is the most likely cause?
2. Give a cross-field validity rule for an e-commerce session table.
3. Why must standardisation precede a frequency table?
4. Which quality dimension does a stale exchange-rate column violate?

LECTURE L17-18 • MODULE 3 • CO3

Missing data

Week 9 | Slides: DATA209_Module_3_Data_Quality_Engineering.pptx, pages 12–21 | Laboratory: P17-18

TOPICS COVERED

- MCAR, MAR, MNAR
- Detection methods including disguised missingness
- Imputation: mean/median, KNN, regression-based

Missingness is data

The pattern of what is absent often carries more information than the values would have. Before imputing anything, ask **why** the value is missing — the answer determines whether any imputation is legitimate.

Three mechanisms

- **MCAR — missing completely at random.** Absence is unrelated to anything. A tube was dropped.

Deletion is unbiased but wasteful; imputation is safe.

- **MAR — missing at random.** Absence depends on *observed* variables. Older patients skipped a test. Imputation conditioned on those variables works.

- **MNAR — missing not at random.** Absence depends on the *unobserved value itself*. High earners hide income. No imputation is unbiased; the missingness must be modelled.

You cannot prove MCAR, nor distinguish MAR from MNAR, from the data alone. You argue for a mechanism from domain knowledge.

Evidence you can gather

Create a binary indicator for missingness in each column and explore it like any other variable. Correlate it with the observed columns — a strong relationship argues against MCAR. Compare the distribution of other variables between the missing and non-missing groups. Visualise the missingness matrix: block patterns reveal structural causes such as a form change. Little's test gives formal evidence against MCAR, but not a MAR/MNAR distinction.

Best of all: **ask the data owner why the field is blank**. This beats every statistical test.

Keep the indicator. It costs one column, preserves the information you are about to overwrite, and documents your assumption.

Disguised missingness

`pima.csv` reports zero nulls. Yet 374 records (48.7%) have Insulin recorded as 0, and 227 (29.6%) have SkinThickness as 0 — and a living person has neither. Sentinel values defeat every automatic null check. This is the single most important idea in the module.

The menu, and its costs

STRATEGY	USE WHEN	COST
Listwise deletion	MCAR, few rows affected	Pima loses 49% of rows to Insulin alone
Column deletion	extreme missingness, weak column	loses a possibly predictive feature
Mean / median	quick baseline, low missingness	shrinks variance, distorts correlations
KNN	MAR, features genuinely related	needs scaling; slow on large data
Iterative / MICE	MAR, several columns affected	complex; can leak if fitted on all data
Indicator + fill	missingness itself is informative	one extra column each

What imputation costs

Mean imputation gives every missing row an identical value, so those rows contribute **zero deviation** from the mean and the standard deviation must fall. In the controlled experiment in P17-18 — where values are deleted with the truth retained — mean imputation lands 1,094 units from the truth on average and shrinks the standard deviation by 6.5%; KNN lands 517 units away and shrinks it by 1.2%. The dataset looks more precise than it is.

Rules

Fit the imputer on the **training data only**, after the split, inside a pipeline. Compare distributions before and after and report the change in mean, standard deviation and skew. Try at least two strategies. State the assumed mechanism in writing: "*median imputation, assuming MAR given Age and BMI*" is a complete claim.

KEY TERMS

MCAR	MAR	MNAR	Disguised missingness	Sentinel value	Listwise deletion
Missingness indicator	KNN imputation	MICE	Little's test		

COMMON ERRORS TO AVOID

- Trusting `isna()` to find all missing values
- Imputing before the train/test split
- Reporting an imputation without measuring the change in variance
- Choosing a strategy without stating the assumed mechanism

READINGS — ALL FREE TO ACCESS

1. pandas — working with missing data
https://pandas.pydata.org/docs/user_guide/missing_data.html
2. scikit-learn — imputation of missing values
<https://scikit-learn.org/stable/modules/impute.html>
3. Kuhn & Johnson — Feature Engineering and Selection
<http://www.feats.engineering/>

TEST YOURSELF

1. Income is missing more often for high earners. Which mechanism, and what follows?
2. Why does mean imputation shrink the standard deviation?
3. Name two pieces of evidence that argue against MCAR.
4. Why must the imputer be fitted after the split rather than before?

LECTURE L19-20 • MODULE 3 • CO3

Outliers

Week 10 | Slides: DATA209_Module_3_Data_Quality_Engineering.pptx, pages 22–30 | Laboratory: P19-20

TOPICS COVERED

- Z-score and IQR methods
- Influence of outliers on modelling
- When an outlier is an error and when it is the finding

What an outlier is

A point far from the bulk of the data **by some stated rule** — a flag raised by a threshold you chose. Sometimes it is an error: a typo, a unit slip, a sentinel. Sometimes it is the most valuable record in the dataset. Fraud detection, fault diagnosis and rare-disease screening are fields where the outliers *are* the entire point.

An outlier is not automatically wrong and not automatically deletable. "Outlier" without a stated threshold is not a claim.

Univariate rules

METHOD	RULE	ASSUMES	FAILS WHEN		
Z-score	\	$x - \text{mean}$ \	$/ \text{SD} > 3$	roughly normal	outliers inflate the SD and mask themselves
Modified z (MAD)	$0.6745 \cdot$ \	$x - \text{median}$ \	$/ \text{MAD} > 3.5$	symmetric	heavily skewed data
IQR fence	outside $Q1 - 1.5 \cdot \text{IQR}$ or $Q3 + 1.5 \cdot \text{IQR}$	nothing	skewed data — over-flags the tail		
Percentile	beyond the 1st or 99th	nothing	always flags 2%, whether or not any exist		

Masking is the trap: with two extreme values present, the standard deviation grows enough that neither exceeds 3. Use the MAD version, which is built on the median.

On raw `pima.csv` the IQR fence flags 34 Insulin points and 29 DiabetesPedigreeFunction points — the two most skewed columns. The fence flags the tail of a skewed distribution *by construction*. That is a signal to transform, not to delete.

Multivariate detection

A 1.9 m height and a 45 kg weight are each unremarkable; together they are worth checking.

- **Mahalanobis distance** — distance from the centre accounting for covariance.

- **Isolation Forest** — random splits isolate anomalies in fewer cuts; scales well.
- **Local Outlier Factor** — compares local density to neighbours'; finds region-relative outliers.
- **Elliptic Envelope** — fits a robust Gaussian; good when the data really is elliptical.

Influence

Outliers move the mean and inflate the standard deviation. They dominate squared-error objectives, so linear regression, k-means and PCA all chase them. High-leverage points can rotate a fitted line on their own. They defeat min-max scaling by stretching the range. Medians, IQRs, rank correlations and tree-based methods are comparatively unbothered.

The decision

Investigate first — always. Check against domain limits and the source record; most decisions become obvious once you know what the value is. Then:

- **Remove** only when demonstrably an error. Record the count and the reason.
- **Treat** — winsorise, cap at a percentile, or transform. Keeps the row, limits the influence.
- **Keep** when it is a genuine extreme. Report it and use robust methods.

The settling question is empirical: **recompute your headline statistic with and without the flagged points.** If it barely moves, stop worrying about them. If it moves a lot, you have found something that needs explaining. Silent deletion is the failure mode.

KEY TERMS

Outlier	Z-score	Median absolute deviation	Masking	IQR fence	Mahalanobis distance
Isolation Forest	Local Outlier Factor	Leverage	Winsorising		

COMMON ERRORS TO AVOID

-
- Deleting flagged points without investigating them
 - Using the z-score rule on skewed or contaminated data
 - Treating IQR flags on a skewed column as errors rather than as a transform signal
 - Failing to report how many rows were removed and why

READINGS — ALL FREE TO ACCESS

-
1. scikit-learn — novelty and outlier detection
https://scikit-learn.org/stable/modules/outlier_detection.html
 2. NIST/SEMATECH — outlier detection
<https://www.itl.nist.gov/div898/handbook/eda/eda.htm>
 3. Kuhn & Johnson — Feature Engineering and Selection
<http://www.feat.engineering/>

TEST YOURSELF

1. Explain masking, and name the rule that avoids it.
2. The IQR fence flags 12% of a column. What should you suspect first?
3. Give a case where the outliers are the object of study.
4. What single test decides whether an outlier decision matters?

LECTURE L21-22 • MODULE 3 • CO3

Encoding strategies

Week 11 | Slides: DATA209_Module_3_Data_Quality_Engineering.pptx, pages 31–37 | Laboratory: **P21-22**
(Assignment 2)

TOPICS COVERED

- One-hot, label and target encoding
- Handling imbalanced and multi-class encoding
- Dummy variables and the dummy variable trap

Why encoding is a modelling decision

Most algorithms require numeric input, and the encoding you choose decides what the model can learn. Assigning 1, 2, 3 to unordered categories invents an order *and* a spacing that do not exist; the model will treat category 3 as "more" than 1 and as equidistant from 2.

For genuinely ordinal data, an ordered integer mapping is correct. For nominal data with few levels, one-hot is the safe default.

The schemes

SCHEME	WHAT IT DOES	USE WHEN	RISK
Label	one integer per level	ordinal data, or tree models	invents order on nominal data
Ordinal (mapped)	integers in a stated order	true ordinal, order known	assumes equal spacing
One-hot	one binary column per level	nominal, low cardinality	column explosion; dummy trap
Binary / hashing	compact numeric encoding	high cardinality	collisions; loses interpretability
Frequency	replace level with its share	high cardinality, frequency meaningful	unrelated levels share a value
Target / mean	replace level with mean target	high cardinality, predictive levels	leakage

Start with one-hot. Move to something cleverer only when cardinality forces you, and then be explicit about the risk.

Two traps

The dummy variable trap. One-hot on k levels produces k perfectly collinear columns — any one is exactly one minus the sum of the others. Linear models cannot invert the resulting matrix, so coefficients become unstable or

undefined. Fix: drop one level as the reference (`drop_first=True`). Trees and regularised models are unaffected, but the redundancy remains.

Target encoding leakage. Replacing a level with its mean target uses the answer to build the feature. Computed on the full dataset, every row sees its own target: validation looks excellent and live performance collapses. Fix: compute the mapping inside cross-validation folds, on training data only, and add smoothing so rare levels are pulled toward the global mean.

If a feature seems implausibly predictive, suspect leakage before celebrating.

Harder cases

High cardinality — group rare levels into "Other" by a stated frequency threshold before encoding. Keep the threshold as a parameter and report it; "levels under 1% of rows" is reproducible. **Unseen levels** at prediction time break a naive encoder — set `handle_unknown='ignore'`.

Imbalance changes evaluation, not encoding, but note it here. At 84.5% / 15.5%, accuracy is not an honest metric: report precision, recall, F1 and PR-AUC, and choose by which error costs more. Stratify every split and every cross-validation fold. Resampling — oversampling, undersampling, SMOTE — belongs *inside the training fold only*; applied before the split it duplicates minority rows across train and test, which is leakage. Class weights are often simpler and safer.

Multi-class targets — check that every class survives the split and that rare classes have enough rows to learn from.

KEY TERMS

One-hot encoding	Label encoding	Ordinal encoding	Target encoding	Dummy variable trap
Reference level	High cardinality	handle_unknown	Stratified split	SMOTE

COMMON ERRORS TO AVOID

- Label-encoding a nominal variable for a linear model
- One-hot encoding without dropping a reference level in a linear model
- Computing target encoding on the full dataset
- Resampling before the train/test split

READINGS — ALL FREE TO ACCESS

1. scikit-learn — preprocessing and encoding
<https://scikit-learn.org/stable/modules/preprocessing.html>
2. scikit-learn — common pitfalls and data leakage
https://scikit-learn.org/stable/common_pitfalls.html
3. Kuhn & Johnson — Feature Engineering and Selection
<http://www.feat.engineering/>

TEST YOURSELF

1. Why does one-hot on k levels break an unpenalised linear model?
2. Describe the correct way to compute target encoding.
3. A categorical column has 4,000 levels. Give two viable strategies.
4. Why is resampling before the split a form of leakage?

LECTURE L23-24 • MODULE 4 • CO4

Transformation and scaling

Week 12 | Slides: DATA209_Module_4_Feature_Engineering_and_Structure.pptx, pages 3-11 | Laboratory: P23-24

TOPICS COVERED

- Log transform, Box-Cox
- Standardisation, MinMax, robust scaling
- Impact of transformation on the distribution

Why transform? Four reasons

1. **Reduce skew** — pull in a long tail so the bulk of the data occupies more of the plot and of the statistic.

1. **Stabilise variance** — when spread grows with the mean, a log transform makes it roughly constant.

2. **Linearise** — a multiplicative relationship becomes additive under logs.

3. **Tame outliers** — compressing the tail reduces influence without deleting any row.

Transformation changes the units. Every interpretation afterwards is in transformed space — say so when you report.

The menu

TRANSFORM	EFFECT	REQUIRES
Log / log1p	strong pull-in of a right tail	strictly positive; log1p allows zero
Square root	milder than log	non-negative; good for counts
Reciprocal	very strong; reverses order	non-zero; rarely appropriate
Box-Cox	searches for the best power λ	strictly positive; $\lambda = 0$ is the log
Yeo-Johnson	Box-Cox generalised	handles zero and negatives
Rank / quantile	forces a chosen distribution	nothing — and is largely uninterpretable

Apply, then **re-measure**. A transform that does not reduce skew has not helped. Students routinely apply one and never check.

Transforming versus scaling

Transformation changes the *shape* (non-linear). **Scaling** changes *location and spread* (linear). Skew survives scaling completely untouched — so transform first if shape is the problem.

Order of operations: **impute** – **transform** – **scale**, and all of it after the split.

Four scalers

SCALER	METHOD	OUTPUT	BEST WHEN
StandardScaler	subtract mean, divide by SD	mean 0, SD 1	roughly symmetric; PCA, linear models
MinMaxScaler	rescale to a fixed interval	usually 0 to 1	bounded input needed, no extreme outliers
RobustScaler	subtract median, divide by IQR	unbounded	outliers present and being kept
MaxAbsScaler	divide by largest absolute value	–1 to 1	sparse data; preserves zeros

One extreme value ruins MinMaxScaler: the range stretches and every ordinary value compresses toward zero.

Does the model care?

METHOD	NEEDS SCALING	WHY
k-NN, k-means	yes	distance-based; a large-range variable dominates
PCA	yes	maximises variance — unscaled, one variable takes PC1 alone
SVM, neural networks	yes	convergence and regularisation depend on scale
Ridge, Lasso	yes	the penalty applies to coefficients, whose size depends on scale
Linear regression (unpenalised)	no	coefficients absorb scale
Trees, random forest, boosting	no	splits are threshold-based, invariant to monotone rescaling

When in doubt, scale. The cost is one pipeline step.

KEY TERMS

Log transform	Box–Cox	Yeo–Johnson	Variance stabilisation	StandardScaler
MinMaxScaler	RobustScaler	MaxAbsScaler	Scale invariance	

COMMON ERRORS TO AVOID

- Scaling a skewed variable and expecting the skew to change
- Using MinMaxScaler on data with extreme values
- Applying Box–Cox to a column containing zeros
- Not re-measuring skew after transforming

1. scikit-learn — comparing the effect of different scalers

https://scikit-learn.org/stable/auto_examples/preprocessing/plot_all_scaling.html

2. scikit-learn — preprocessing

<https://scikit-learn.org/stable/modules/preprocessing.html>

3. Kuhn & Johnson — Feature Engineering and Selection

<http://www.feat.engineering/>

TEST YOURSELF

1. A column has skew 2.3 and contains zeros. Which transform, and why not Box-Cox?
2. Why does scaling leave skewness unchanged?
3. Which of random forest and k-NN needs scaled input, and why?
4. Name the correct order of impute, scale, transform and split.

LECTURE L25-26 • MODULE 4 • CO4

Feature engineering and selection

Week 13 | Slides: DATA209_Module_4_Feature_Engineering_and_Structure.pptx, pages 12-20 | Laboratory: P25-26

TOPICS COVERED

- Feature construction; interaction and polynomial features
- Curse of dimensionality
- Feature selection: filter, wrapper, embedded

A feature is a question

Feature engineering is where domain knowledge enters the pipeline. It has the highest return on effort of any step and is the one most often skipped.

Six ways to build one:

- **Ratios** — BMI from weight and height; conversion rate from clicks and views.
- **Differences** — days since last visit; deviation from a group mean.
- **Aggregates** — per-entity mean, max or count.
- **Date parts** — month, weekday, is-weekend, hour. One date becomes several features.
- **Binning** — age into clinical bands. Loses information, gains robustness.
- **Indicators** — `was_missing`, is-zero, exceeds-threshold. Cheap, interpretable, often strong.

Every constructed feature needs a one-line rationale. If you cannot say why it should matter, do not build it.

Interactions and polynomials

Interaction features are products of two variables, capturing effects neither shows alone. Linear models cannot discover these; trees can, partially. **Polynomial features** are powers of a variable, letting a linear model fit a curve.

`PolynomialFeatures(degree=2)` on 8 columns yields 44 — count before you generate. Both overfit quickly; pair with regularisation and validation.

Generate deliberately, not exhaustively. Automatic expansion is the fastest route into the curse of dimensionality.

Three families of selection

- **Filter** — score each feature independently of any model (correlation, chi-square, mutual information, variance). Fast, model-agnostic, blind to interactions.

- **Wrapper** — search subsets by training a model on each (RFE, forward/backward selection).

Accurate for that model, expensive, overfits easily.

- **Embedded** — selection happens during fitting: Lasso shrinks coefficients to zero, trees report

importances. A good default balance.

	FILTER	WRAPPER	EMBEDDED
Cost	very low	high	moderate
Model-specific	no	yes	yes
Sees interactions	no	yes	partly
Overfitting risk	low	high	moderate

Run a filter first to remove the obviously useless, then an embedded method on what survives. Reserve wrappers for small feature sets.

Agreement is the signal

Applying five methods to the shoppers feature set, `PageValues`, `has_page_value` and `ExitRates` are chosen by all five; several are chosen by only one. **A feature selected by one method alone deserves scrutiny.**

The honest test of feature engineering is whether the constructed features survive selection — and whether a smaller set performs comparably. On the shoppers data, 18 features give cross-validated ROC-AUC 0.906 and a 6-feature consensus set gives 0.875. The smaller set is cheaper, more stable and easier to explain; whether the difference is worth it is a judgement you must state.

Select **inside** cross-validation. Selecting on the full dataset and then validating is leakage.

KEY TERMS

Feature construction	Interaction feature	Polynomial feature	Binning	Filter method
Wrapper method	Embedded method	RFE	Mutual information	Lasso

COMMON ERRORS TO AVOID

- Generating polynomial features exhaustively without counting them
- Selecting features on the full dataset before splitting
- Trusting a single selection method's ranking
- Building features with no stated rationale

READINGS — ALL FREE TO ACCESS

1. scikit-learn — feature selection
https://scikit-learn.org/stable/modules/feature_selection.html
2. Kuhn & Johnson — Feature Engineering and Selection
<http://www.feat.engineering/>
3. Kaggle Learn — Feature Engineering
<https://www.kaggle.com/learn/feature-engineering>

TEST YOURSELF

1. Give three features you could derive from a single timestamp column.
2. Why can a filter method miss a genuinely useful feature?
3. You have 40 features and 300 rows. Which selection family, and why?
4. Why must feature selection sit inside cross-validation?

LECTURE L27-28 • MODULE 4 • CO4

Dimensionality reduction

Week 14 | Slides: DATA209_Module_4_Feature_Engineering_and_Structure.pptx, pages 21-28 | Laboratory: P27-28

TOPICS COVERED

- PCA — interpretation focused
- LDA — the classification view
- t-SNE and UMAP for visualisation only
- Interpreting components; multicollinearity

Selection versus reduction

Selection keeps a subset of the original columns; the output stays interpretable and the information in dropped columns is discarded. **Reduction** builds new columns as combinations of the originals; interpretation is harder but information from every column is retained.

Prefer selection when interpretability is the deliverable, reduction when variables are collinear or you need a 2-D view. Say which you did and why.

PCA in five steps

1. **Standardise** — centre and scale. Skipping this makes PCA measure units, not structure.
2. **Covariance** — compute the covariance matrix of the standardised features.
3. **Decompose** — find eigenvectors (directions of variance) and eigenvalues (how much lies along each).
4. **Order and cut** — sort by eigenvalue; keep as many as your variance target requires.
5. **Project** — transform onto the retained components, each orthogonal to all the others.

PCA is **unsupervised**: it never looks at the target, so the direction of greatest variance need not be the useful direction.

Reading the output

Retention rules: a cumulative variance target (commonly 80–95%), the scree elbow, or the Kaiser rule (eigenvalue > 1). State which you used.

On the transformed shoppers features, PC1 explains 47.9% and four components reach 79.7% — a meaningful compression. On standardised `pima.csv`, the profile is 28.5 / 18.7 / 14.3 / 11.5 ..., reaching 80% only at the fifth component. **A flat scree plot is a finding**, not a failure: it means the variables were already fairly independent.

Interpreting components

Read the **loadings** — the weight of each original variable in the component. A component with large weights on Glucose, Insulin and BMI is a "metabolic load" axis; name it. Signs matter: opposite signs mean the component contrasts two groups of variables.

A component you cannot name is a component you should not present to a domain audience.

Cautions: scale first; PCA is linear only; it is not appropriate for one-hot encoded categories, whose binary geometry breaks its assumptions; outliers distort components; and it ignores the target.

Multicollinearity detection with VIF

Reduction and multicollinearity are two responses to the same problem, so check for it before you reduce. The **variance inflation factor** for a predictor is $1 / (1 - R^2)$ from regressing that predictor on all the others: above 5 it is worth investigating, above 10 the column is largely reconstructable from the rest. (Introduced in L11-12; applied here as a screening step.)

Three responses, in increasing order of cost to interpretability:

1. **Drop** one of the pair — cheapest, and defensible when the two measure the same thing.
2. **Combine** them into a single constructed feature — keeps the information, keeps a name.
3. **Reduce** with PCA — keeps all the information and produces orthogonal components by

construction, so multicollinearity is eliminated rather than managed. The cost is that the components need naming before anyone can act on them.

Note the asymmetry: PCA *removes* multicollinearity but does not tell you which original variable was redundant. If the deliverable is a recommendation about which measurements to keep collecting, use VIF and selection instead.

Alternatives

METHOD	SUPERVISED	OPTIMISES	USE FOR
PCA	no	maximum retained variance	compression, decorrelation, interpretation
LDA	yes	maximum class separation	reduction with a label; at most $k-1$ components
t-SNE	no	local neighbourhood preservation	2-D visualisation only
UMAP	no	local and some global structure	2-D visualisation; faster, more stable

t-SNE and UMAP traps: cluster sizes and between-cluster distances are not meaningful, and results change with the seed and with perplexity. Never use their output as model input.

Finally: PCA belongs **inside** the pipeline, refitted on each training fold. Fitting it on the whole dataset before cross-validation leaks test information.

KEY TERMS

Principal component analysis	Eigenvector	Eigenvalue	Explained variance ratio		
Scree plot	Loading	Kaiser rule	LDA	t-SNE	UMAP

COMMON ERRORS TO AVOID

- Running PCA on unstandardised features
- Applying PCA to one-hot encoded categorical columns
- Reading cluster sizes or distances off a t-SNE plot
- Fitting PCA on the full dataset before cross-validation

READINGS — ALL FREE TO ACCESS

1. scikit-learn — decomposition and PCA
<https://scikit-learn.org/stable/modules/decomposition.html>
2. Stanford CS229 — lecture notes
<https://cs229.stanford.edu/>
3. James et al. — Introduction to Statistical Learning, Ch. 12
<https://www.statlearning.com/>

TEST YOURSELF

1. Why must features be standardised before PCA?
2. PC1 explains 12% of variance. What does that tell you about the data?
3. Why is t-SNE unsuitable as a preprocessing step for a classifier?
4. How do you turn a loading vector into something a domain expert can use?

LECTURE L29-30 • MODULE 4 • CO4

Closing the loop

Week 15 | Slides: DATA209_Module_4_Feature_Engineering_and_Structure.pptx, pages 29-37 | Laboratory: P29-30

TOPICS COVERED

- Hypothesis refinement; train/test splitting
- Cross-validation overview
- Data leakage: concept and prevention
- Translating EDA into insight

Splitting

The test set exists to estimate performance on data the process has never seen. **Touch it once.**

- Split **before** imputing, scaling, encoding, selecting or transforming — before everything.
- **Stratify** on the target when classes are imbalanced.
- **Time-ordered data** requires a time-based split: train on the past, test on the future, never shuffle.
- **Grouped data** must be split by group, so all records for one entity sit on one side.
- Set and record a random seed so the marker reproduces exactly what you saw.

Cross-validation

One split gives one noisy estimate. k-fold splits into k parts, training on k-1 and testing on the remainder, k times; k = 5 or 10 is standard. **Stratified k-fold** preserves class proportions and is the default for imbalanced classification. **Time-series split** uses expanding or rolling windows. **Group k-fold** keeps entities together.

Report the mean **and the spread** across folds. A high mean with a wide spread is an unstable result, not a good one.

Data leakage

Leakage is any path by which information from the test set — or from the future — reaches the training process.

It is the most expensive error in applied data science because it announces itself as *excellent* validation performance and only fails in production.

PATH	RISK	FIX
Preprocessing on all data	scaler or imputer sees test statistics	fit inside a Pipeline , on train only
Target leakage	a feature contains the answer	audit each feature's availability at prediction time
Temporal leakage	training on rows recorded after test rows	split by time; check both date ranges
Duplicate leakage	the same entity in train and test	deduplicate first, then split by group

The diagnostic question for every feature: *would this value have been available, with this content, at the moment of prediction?* A useful blunt check is single-feature AUC — anything above about 0.95 suggests the feature is standing in for the target.

Sampling skewed data — imbalanced targets

Identify the imbalance during EDA and state it. Accuracy is not honest at 85/15. Report precision, recall, F1 and PR-AUC, and choose by which error costs more. Stratify every split and every fold. Resampling belongs inside the training fold only; class weights are simpler and leak nothing.

Turning exploration into insight

The structure of a credible EDA report, and of your final project:

1. **What the data is** — source, sampling frame, rows, columns, period; what it cannot speak to.
2. **What was wrong with it** — the cleaning log, and every decision you made.
3. **What it shows** — findings, each with the figure that supports it and a sentence of meaning.
4. **What comes next** — hypotheses with falsification conditions, the modelling-ready dataset,

and the caveats.

A negative result, well established, is a contribution. "K-means finds no strong segmentation (silhouette 0.20)" is a finding worth reporting.

Review — the semester in one page

MODULE	THE QUESTION IT ANSWERS	THE ARTEFACT IT PRODUCES
1 · Foundations	what is this data, and what may I claim from it?	a described dataset with its sampling frame stated
2 · Core EDA	what patterns and structure does it contain?	findings and a hypothesis shortlist
3 · Data quality	can I trust it?	a cleaning log and a reliable analytical dataset
4 · Structure	what shape should it take for modelling?	a leak-free, modelling-ready feature matrix

Five ideas that recur in every module, and that the final project is really marking:

1. **Look before you test.** Compute the summary, then plot it, then reconcile the two.
2. **Every choice is an assumption.** Imputing, deleting, transforming and encoding all assert something. Write the assertion down.

1. **Match the statistic to the variable.** Scale decides the statistic; shape decides the summary.

2. **Measure the effect of your own decisions.** Before-and-after, every time.

3. **Say what the data cannot support.** The limitations section is part of the finding, not an apology for it.

KEY TERMS

Train/test split

Stratification

k-fold cross-validation

Group k-fold

Time-series split

Data leakage

Target leakage

Temporal leakage

PR-AUC

Pipeline

COMMON ERRORS TO AVOID

- Fitting any transformer before the split
- Reporting accuracy on an imbalanced target
- Evaluating on the test set more than once
- Resampling or selecting features outside the cross-validation loop

READINGS — ALL FREE TO ACCESS

1. scikit-learn — common pitfalls and data leakage

https://scikit-learn.org/stable/common_pitfalls.html

2. scikit-learn — cross-validation

https://scikit-learn.org/stable/modules/cross_validation.html

3. Berkeley Data 100 — course notes

<https://ds100.org/course-notes/>

TEST YOURSELF

1. Name the four leakage paths and one fix for each.
2. Why is a random split wrong for a dataset with timestamps?
3. A feature alone gives AUC 0.98. What is your first suspicion?
4. Give the four-part structure of an EDA report.